$$y_n \sim Normal(\mu_n, \sigma_{Residual}) \tag{7}$$

$$\mu_n = \beta_{(Intercept)} + u_{subj[n],1} + \tag{8}$$

$$(\beta_{meA} + u_{subj[n],2}) \times \text{meA}_n + \tag{9}$$

$$(\beta_{meB} + u_{subj[n],3}) \times \text{meB}_n + \tag{10}$$

$$(\beta_{int} + u_{subj[n],4}) \times \text{int}_n \tag{11}$$

The random effects u are multivariate normally distributed:

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \Sigma_u\right), \tag{12}$$

where Σ_u is the variance-covariance matrix, with the standard deviation parameters $\sigma_{Random\,slopes}$ and the correlation parameters $\rho_{Random\,slopes}$. The contrast coding for the two main effects and the interaction (meA$_n$, meB$_n$, and int$_n$) was set to sum-coding ($-1/+1$). The model parameters used for simulating the SBC data were inspired by our own data on the task (Schad et al., 2014). The prior distributions (from which simulating parameter values were sampled) were the following:

$$\beta_{(Intercept)} \sim Normal(0.7, 0.1) \tag{13}$$

$$\beta_{Fixed\,effects} \sim Normal(0, 0.1) \tag{14}$$

$$\sigma_{Random\,slopes} \sim Normal_+(0, 0.1) \tag{15}$$

$$\sigma_{Residual} \sim Normal_+(0, 0.1) \tag{16}$$

$$\rho_{Random\,slopes} \sim LKJ(2) \tag{17}$$

where $Normal_+$ denotes a normal distribution truncated at zero.

We performed three SBC simulations, one for main effect A, one for main effect B, and one for the interaction A $\times$ B. Each simulation used the same H1 model (shown above). In each simulation, the null model (H0) was created by setting one fixed effects parameter to zero. When performing SBC for main effect A, $\beta_{meA} = 0$; when performing SBC for main effect B, $\beta_{meB} = 0$; when performing SBC for the interaction, $\beta_{int} = 0$. In each SBC simulation, we defined prior model probabilities as $p(H0) = 0.5$ and $p(H1) = 0.5$. In each SBC simulation, we fitted the same models to the data that were used to generate the data (i.e., H1 and the corresponding H0). In the model fitting, we used $10,000$ iterations of the MCMC sampler, with $2,000$ iterations of warm-up. For the bridge sampling, we used the "warp3" method. We performed 200 runs of SBC.

Design 2: One 2-level factor with crossed random effects for subjects and items

We simulated data from a Latin square design with one two-level fixed factor and crossed random effects for 42 subjects and 16 items, leading to a total of $42 \times 16 = 672$ data points. We set the prior probability of the alternative hypothesis (H1) to a smaller value of 0.2 (instead of the previous 0.5) to be able to detect potential liberal biases well without ceiling effects. A Bayesian GLMM with a lognormal likelihood was used in the SBC analyses using the package brms. We included correlated random slopes and intercepts for subjects and for items.

$$y_n \sim LogNormal(\mu_n, \sigma_{Residual}) \tag{18}$$

$$\mu_n = \beta_{(Intercept)} + u_{subj[n],1} + w_{item[n],1} + \tag{19}$$

$$(\beta_X + u_{subj[n],2} + w_{item[n],2}) \times X_n \tag{20}$$

The random effects u and w are each multivariate normally distributed:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma_u\right), \tag{21}$$

$$\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma_w\right), \tag{22}$$

where Σ_u and Σ_w are the variance-covariance matrices, with the standard deviation parameters $\sigma_{Random\,slopes}$ and the correlation parameters $\rho_{Random\,slopes}$. The contrast coding for the fixed effect (X_n) was set to sum-coding ($-1/+1$). The prior distributions (from which simulating parameter values were sampled) were the following:

$$\beta_{(Intercept)} \sim Normal(6, 0.6) \tag{23}$$

$$\beta_X \sim Normal(0, 0.15) \tag{24}$$

$$\sigma_{Random\,slopes} \sim Normal_+(0, 0.1) \tag{25}$$

$$\sigma_{Residual} \sim Normal_+(0, 0.5) \tag{26}$$

$$\rho_{Random\,slopes} \sim LKJ(2) \tag{27}$$

We performed one SBC simulation comparing this full model (H1) to a null model (H0), where the fixed effect was set to zero: $\beta_X = 0$. In the SBC, we fitted the same models to the data that were used to generate the data (i.e., H1 and H0). In the model fitting, we used $10,000$ iterations, with $2,000$ iterations of warm-up. For bridge sampling, we again used the method “warp3” with the aim to reduce the number of warning messages.